

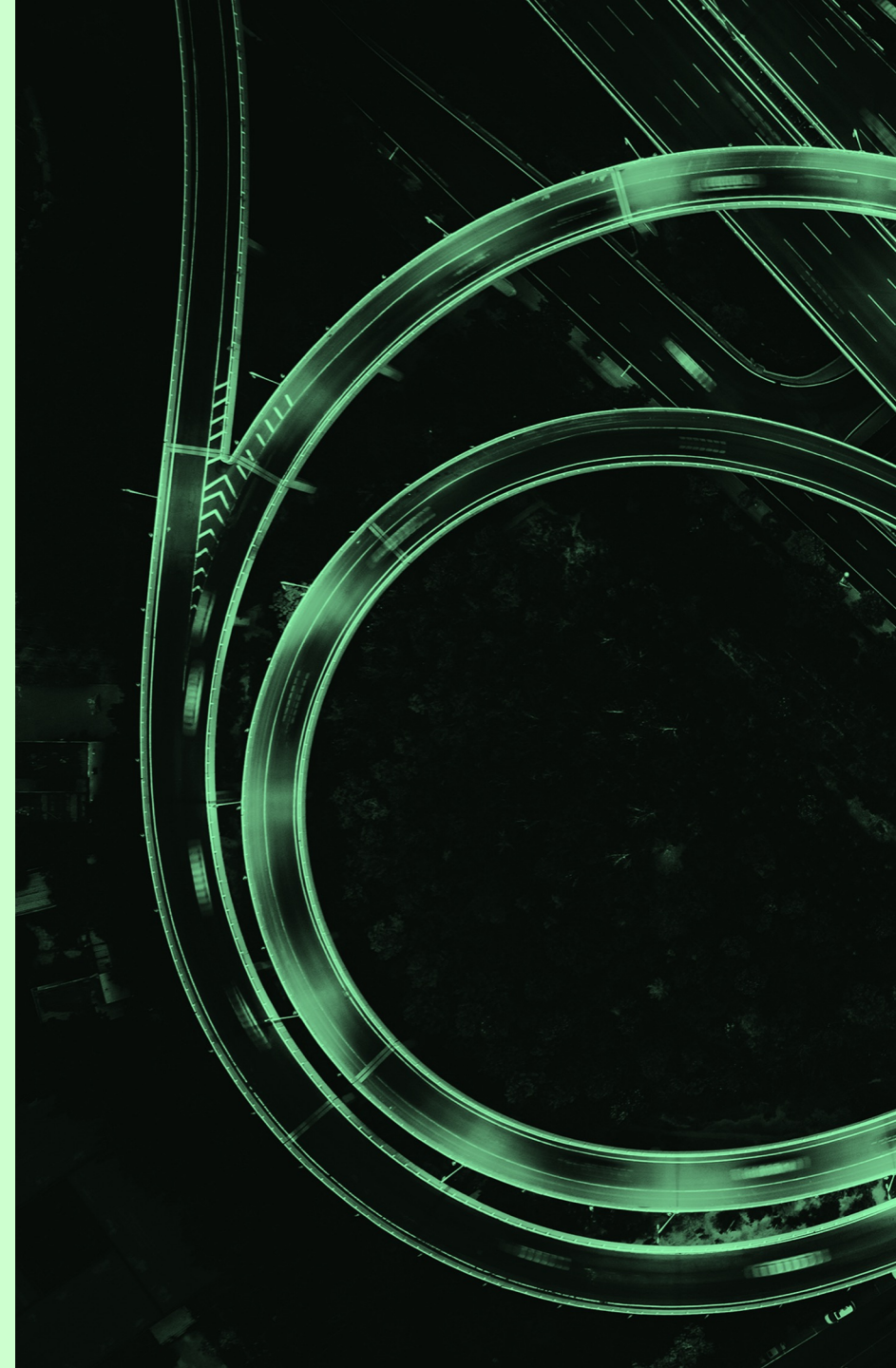
When **Agents** **Run Healthcare**

Building Reliable Agentic Systems in
Highly Regulated Environments

AGNTCon + MCPCon Europe 2026

Janosch Woschitz | Senior Cloud & AI Architect

BARMER



40%

of enterprises will **demote or decommission autonomous AI agents by 2027** due to governance gaps discovered only after production incidents occur.

Gartner, 2026

German Healthcare System

The system, in a nutshell

Public statutory health insurance (***Gesetzliche Krankenversicherung, GKV***) covers the vast majority of the population. A large, tightly regulated, and operationally complex system.

74.2M

people insured under statutory health insurance

>90% of Germany's population

€353B

spent through the GKV system every year

93

self-governed insurers - one legal framework (SGB) sets the boundaries: regulated, but decentralized

Who is BARMER

8.3M

insured people, one of the largest
GKV in Germany

14,000+

employees across Germany

Nationwide

branch network serving members
in every region



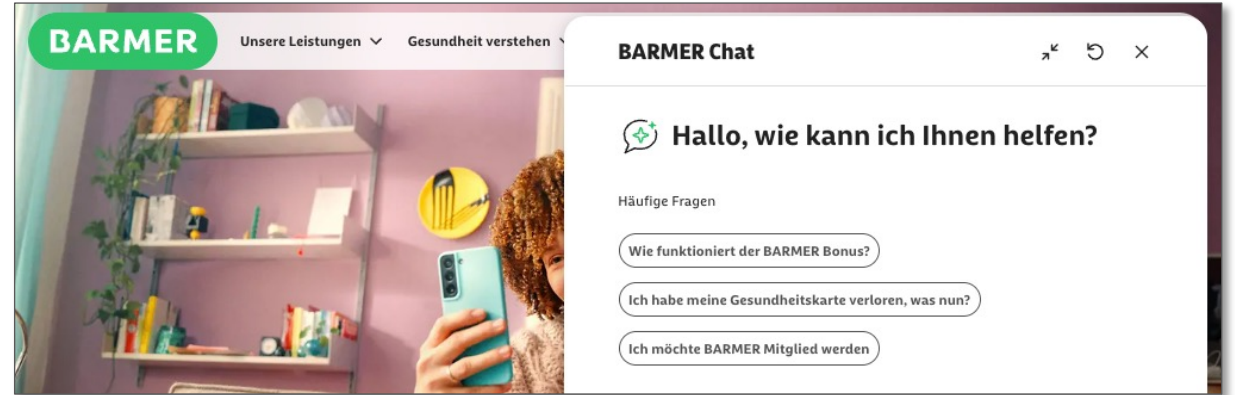
BARMER

From data platforms to agentic workflows

How BARMER.KI evolved from AI foundations to agents



Janosch Woschitz · AI Architect · BARMER.KI



DATA

Data & Analytics

Scalable data foundations,
cloud, governance

DECISIONS

Analytics & ML at scale

Predictive models and
decision support

INTERACTIONS

Conversational AI & assistants

BARMER Chat and
internal assistants

ACTIONS

Agentic workflows

Why **agentic**, why now

Two forces are pulling apart and hiring can't close the gap

Demand

↑ An aging population means more claims, more complex care, more admin per member.

Supply

↓ The workforce is shrinking, while experienced employees are retiring.

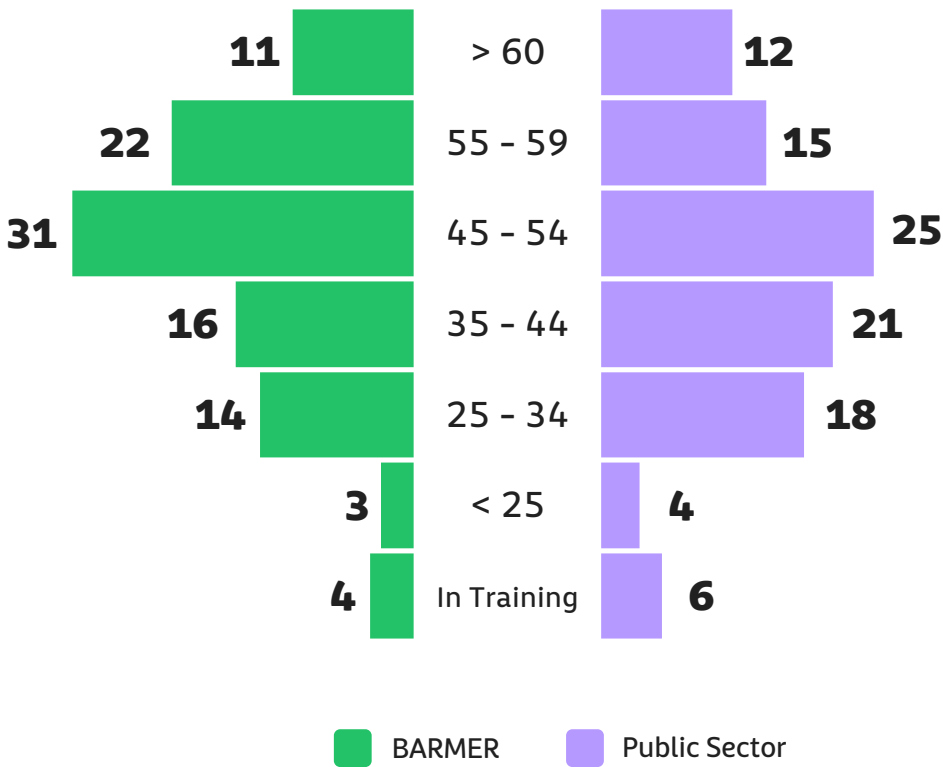
BARMER's response

“Agenten Automatisieren Arbeitsabläufe”

→ Agents automate workflows

Workforce Age Distribution

Employee age distribution, 2025 (%)



When agents enter core processes

Autonomy introduces a different class of engineering challenges

01	02	03	04	05
Probabilistic reasoning Agents operate with uncertainty. Core processes require defensible outcomes.	Autonomy & control More autonomy increases the impact of errors.	Auditability & accountability Consequential actions must remain traceable and attributable.	Sensitive data & access Agents need context, but access must remain purpose-bound and least-privilege.	Core-system integration Agents must work across existing environments without destabilizing core systems.

The challenge isn't making agents intelligent enough. It's making their autonomy **governable**.

And then there is regulation

One agentic workflow. Multiple regulatory regimes. All at once.

GDPR

Personal data

Purpose limitation, lawful processing, and data-subject rights

Related: BDSG

EU AI Act

AI governance

Risk classification, documentation, oversight, and transparency

SGB IV / V / X

Health & social data

Sector-specific rules on what data may be used, by whom, and why

BSI / C5 / ...

Infrastructure & security

Security controls, assurance, cloud, and operational requirements

Related: NIS2 · KRITIS · CADA



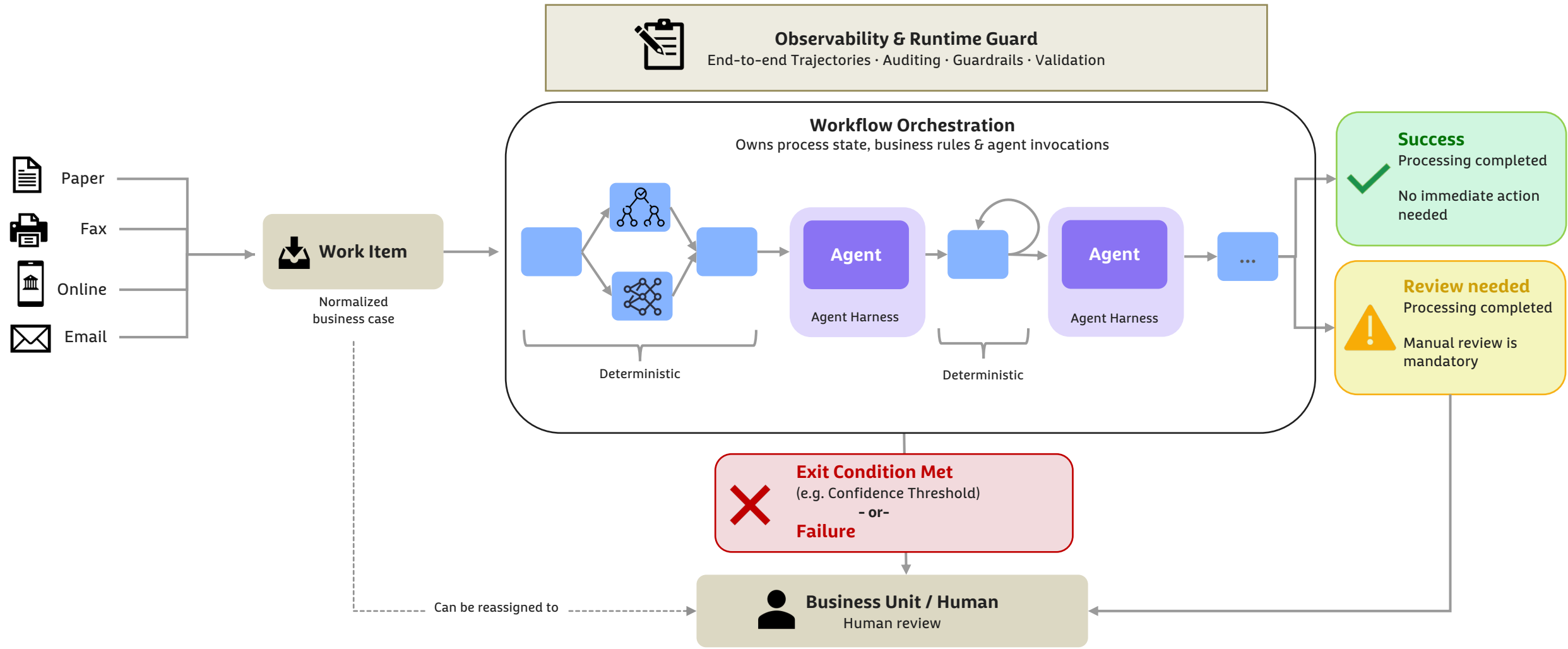
Regulation isn't a layer we add around the agent. It shapes how the system must be designed.

Architecture

The **How**

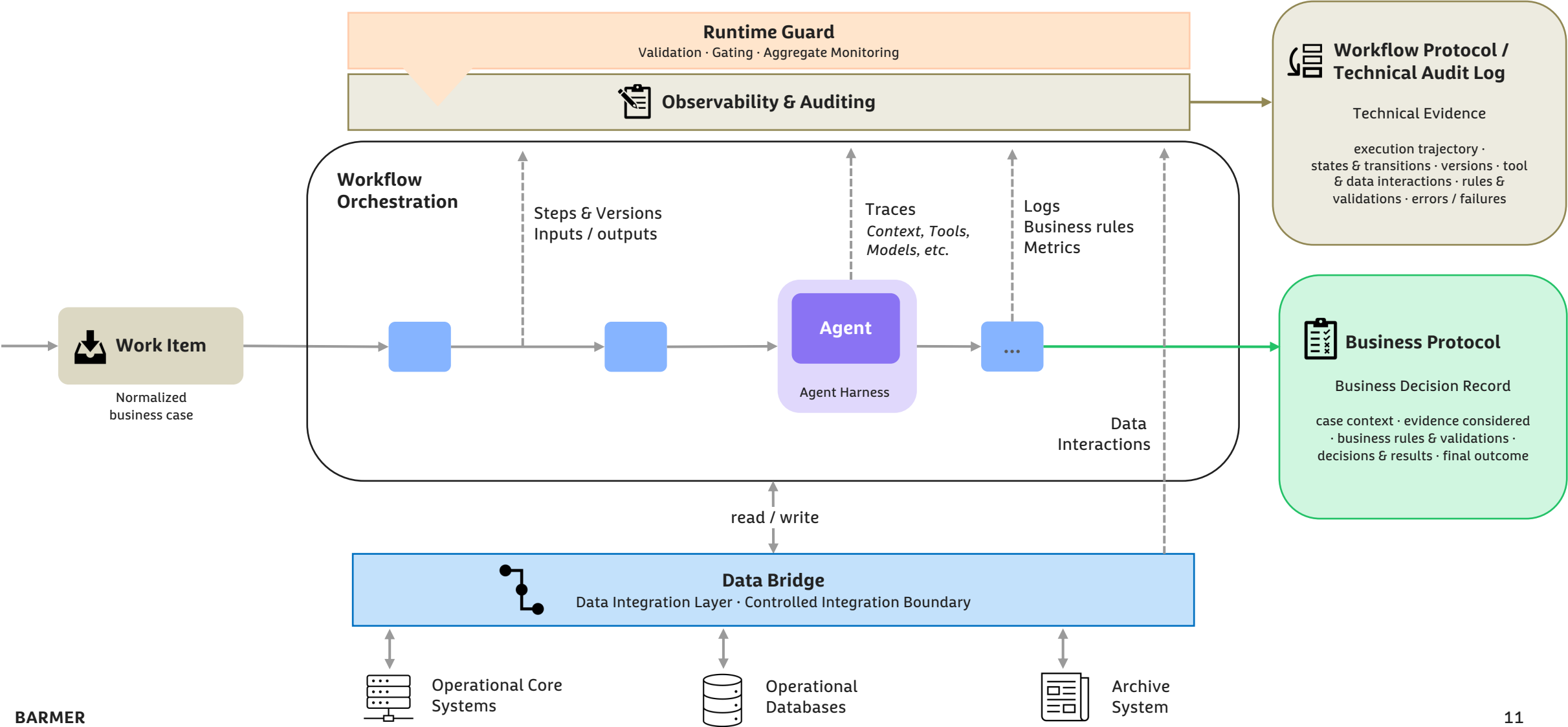
Determinism first. AI where it earns its place.

Primarily deterministic, agentic where necessary, human in oversight



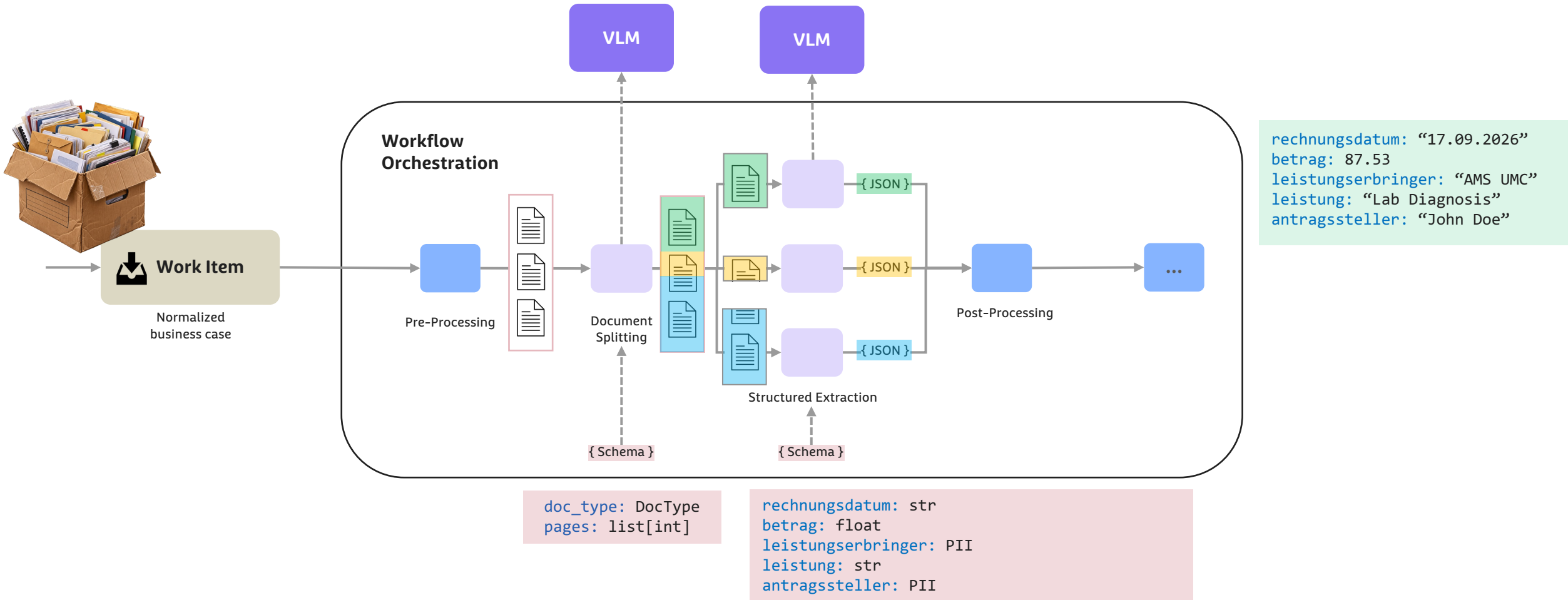
Observe end-to-end. Expose only what is needed.

Durable evidence and controlled data access by design



Model-based understanding. Controlled orchestration.

VLM-based document splitting and structured extraction



Walkthrough

One Case, End to End

Raw input

This is what arrives



1



2



3



4



5



6



7



8



9



10



11



12



13



14



15

A single PDF containing different kinds of documents

Split + classification

Bringing Structure to the Case

[1 – 4]
Claims Form

[5 – 6]
Receipt

[7 – 8]
Receipt

9
Receipt

[10 – 11]
Receipt

12
Receipt

13
Receipt

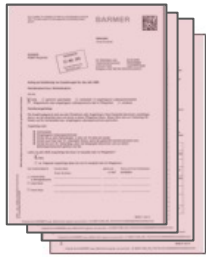
14
Other /
Not relevant

15
Receipt

Documents are split and classified by type

Extraction

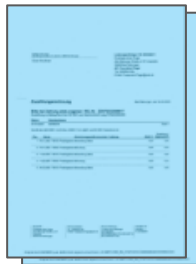
From Documents to Data



[1 - 4]
Claims Form



```
{
  "pages":
  [
    1,
    2,
    3,
    4
  ],
  "reasoning": "BARMER-Antrag auf Erstattung von Zuzahlungen für das Jahr 2025, 4-seitiges Form",
  "doc_type": "Antrag",
  "extracted":
  {
    "versicherungsnummer": "V123456789",
    "antragsdatum": "XX.XX.2026",
    "beantragte_leistungen":
    [
      "Erstattung von Zuzahlungen für das Jahr 2025"
    ],
    "unterschrift_vorhanden": true,
    "chroniker_information": true,
  }
}
```



[5 - 6]
Receipt



```
{
  "pages":
  [
    5,
    6
  ],
  "reasoning": "Zuzahlungsrechnung der Podologie XXXXXXXX, RG.-Nr. ZZZZZZZZZX817, Datum XX.XX.2025, Se",
  "doc_type": "Belege/Quittungen",
  "extracted":
  {
    "belege":
    [
      {
        "beleg_id": "B001",
        "beleg_reasoning": "Zuzahlungsrechnung (Heilmittel/Podologie) - Eigenanteil von 3,42 € ex",
        "rechnungsdatum": "XX.XX.2025",
        "leistung_von": "XX.XX.2025",
        "leistung_bis": "XX.XX.2025",
        "betrag": 3.42,
        "leistungserbringer": "Podologie XXXXXXXX",
        "leistung": "78010 / Podologische Behandlung (klein)",
      }
    ]
  }
}
```

Structured extraction turns relevant content into machine-readable data

Business protocol

What Happened?

Protokoll aus Antragsverarbeitung AI

Dokumentation

Verarbeitungsergebnis

Technisches Ergebnis

Erfolg

Verarbeitung erfolgreich abgeschlossen

Fachliches Ergebnis

Erstattung: 22,35 €

Berücksichtigte Zuzahlungen: 190,12 €

Belastungsgrenze: 167,77 €

Verfahren

Kundenantrag Befreiung § 62

Fallbündelnummer

██████████0908

Datum Posteingang verarbeiteter Antrag

██████████2026

Bezugsjahr

2025

KVNRDE

██████████

BP-NR

██████████

Name des Versicherten

Ldrei Wfünfzwei

Datum Verarbeitung durch Agent

11.08.2026 11:56:18

1. aufgetretener Fehler

—

Protokoll aus Antragsverarbeitung AI

Dokumentation

Verarbeitungsergebnis

Technisches Ergebnis

Fehlgeschlagen

1. aufgetretener Fehler in Phase: Datenabgleich (Nachverarbeitung)
Grund: Antrag-Einkunftsart ist ungültig

Fachliches Ergebnis

Keine Entscheidung

Keine Berechnung möglich — Verarbeitung / Prüfung(en) fehlgeschlagen

Verfahren

Kundenantrag Befreiung

Fallbündelnummer

██████████4909

Datum Posteingang verarbeiteter Antrag

██████████2026

Bezugsjahr

2025

KVNRDE

██████████

BP-NR

██████████

Name des Versicherten

Evier Snullvier

Datum Verarbeitung durch Agent

11.08.2026 11:57:55

1. aufgetretener Fehler

EXIT_023 – Antrag-Eink Fallbündel (siehe Detai

Protokoll aus Antragsverarbeitung AI

Dokumentation

Verarbeitungsergebnis

Technisches Ergebnis

Erfolg

Verarbeitung erfolgreich abgeschlossen

Fachliches Ergebnis

Ablehnung

Belastungsgrenze noch nicht erreicht

Verfahren

Kundenantrag Befreiung § 62

Fallbündelnummer

██████████9910

Datum Posteingang verarbeiteter Antrag

██████████2026

Bezugsjahr

2024

KVNRDE

██████████

BP-NR

██████████

Name des Versicherten

Teins von der Kzweiacht

Datum Verarbeitung durch Agent

11.08.2026 11:57:05

1. aufgetretener Fehler

—

Detaillierte Postprocessing-Entscheidungen (Datenabgleich)

R010: Prüft ob Dokumententyp 'Sonstiges' vorhanden ist (nicht zulässig)

Entität

Details

Status

Fallbündel

0 ms

Erfolgreich

R011: Prüft Einkommensdaten gegen Antrag/Nachweise: Übereinstimmung Renteneinkunftsart zwischen Antrag und SAP, nur zulässige Einkunftsarten in Antrag und Einkommensnachweisen

Entität

Details

Status

Fallbündel

Exit: EXIT_023 - Antrag enthält nicht unterstützte Einkommensarten: Einkommen von Angehörigen Vorgefunden: 'Altersrente', 'Einkommen von Angehörigen' | Erwartet: None oder ausschließlich 'Altersrente', 'Erwerbsminderungsrente' oder 'Hinterbliebenenrente'

Nicht erfüllt

R024: Prüft Chroniker-Übereinstimmung: SAP-Status muss mit Antrag und Chronikerbescheinigung übereinstimmen

Entität

Details

Status

Fallbündel

0 ms

Erfolgreich

R012: Prüft alle Belege auf strukturelle Vollständigkeit: Leistungsdatum vorhanden und im Bezugsjahr, Antragssteller stimmt mit Beteiligten überein

Entität

Details

Status

Beleg B001

0 ms

Erfolgreich

Beleg B002

0 ms

Erfolgreich

The business protocol captures the case in business terms

Workflow protocol

How Did It Happen?

```
{
  "execution_id": "28d85065-9c1d-4d9a-a984-85d37cf5cdc0",
  "status": "SUCCEEDED",
  "start_time": "2026-08-20T13:50:42.363000+00:00",
  "stop_time": "2026-08-20T13:54:11.786000+00:00",
  "state": [
    {
      "state_name": "InvokeSapDataFetch",
      "state_type": "Task",
      "input": {
        "data": {
          "fallbuendel_id": "XXXXXXXXXXXX0910",
          "instance_id": "XXXXXXX449",
          "use_mock_sap_data": false,
          "write_charges": false,
          "write_activities": false,
          "write_refund_zuz": false,
          "write_close_fallbuendel": false,
          "write_charge_free": false,
          "forward_workitem": false,
          "write_to_nuxeo": false
        },
        "timestamp": "2026-08-20T13:50:42.429000+00:00"
      },
      "output": {
        "data": {
          "fallbuendel_id": "XXXXXXXXXXXX0910",
          "instance_id": "XXXXXXX449",
          "use_mock_sap_data": false,
          "write_charges": false,
          "write_activities": false,
          "write_refund_zuz": false,
          "write_close_fallbuendel": false,
          "write_charge_free": false,
          "forward_workitem": false,
          "write_to_nuxeo": false,
          "sap_result": {

```



80,000+ lines for a single case

The technical protocol captures the complete execution record

Beyond the Workflow

Building for Production

What we didn't cover

Production engineering extends beyond the runtime architecture



Release Engineering

- Offline evals & software testing (unit / integration / smoke tests)
- Automated quality gates & release processes



Governance & Documentation

- Documentation requirements & curation
- Regulatory assessment of individual use cases & processes



AI-assisted Engineering

- AI-assisted development across the engineering lifecycle
- Faster exploration, implementation & iteration



From agentic to deterministic

- Highly agentic approaches during development
- Harden into deterministic production pathways wherever possible

Key Takeaways

What we learned building agents for production

01

Introduce autonomy selectively

Explicit behaviour makes the system easier to control, test and explain

Deterministic pathways keep the process controllable; use agentic reasoning where it adds value.

02

Promote recurring needs into shared capabilities

Reliability emerges from the system, not any single component

Shared capabilities make access, integration and execution concerns consistent across workflows.

03

Translate regulation into engineering

Turn regulatory requirements into explicit, testable system properties

Express requirements through controls, evidence, access boundaries and escalation paths that can be verified.

Thank You

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BARMER